Exercises

1)

a) $A^T A = \begin{bmatrix} 1 & \cdots & 1 \\ x_0 & \cdots & x_n \end{bmatrix} \begin{bmatrix} 1 & x_0 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n+1 & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix}$

Top left: Total # of data points (N).

Off-diagonals: Related to the mean ($\bar{x} = \frac{1}{N} \Sigma x_i$)

Bottom right: Related to the 2nd moment (and variance)

b) $A^T f = \begin{bmatrix} 1 & \cdots & 1 \\ x_0 & \cdots & x_n \end{bmatrix} \begin{bmatrix} f_0 \\ \vdots \\ f_n \end{bmatrix} = \begin{bmatrix} \Sigma_{i=0}^{n} f_i \\ \Sigma_{i=0}^{n} x_i f_i \end{bmatrix}$

Top: Sum of the dependent variables

Bottom: Sum of products related to the covariance between x and f

c) $A^T A \hat{a} = A^T f$

$S_x = \Sigma x_i \quad S_{xx} = \Sigma x_i^2 \quad S_f = \Sigma f_i \quad S_{xf} = \Sigma x_i f_i$

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \frac{1}{(n+1) S_{xx} - (S_x)^2} \begin{bmatrix} S_{xx} & -S_x \\ -S_x & n+1 \end{bmatrix} \begin{bmatrix} S_f \\ S_{xf} \end{bmatrix}$$

Computer lab

1) $r_1 + r_2 + r_3 + r_4 = 20$ $\qquad$ $Ar = b$

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \qquad b = \begin{bmatrix} 4 \\ 9 \\ 6 \\ 3 \\ 7 \\ 20 \end{bmatrix}$$

$$(A^T A) r = A^T b$$

$r_1 = 4.67$

$r_2 = 1.50$

$r_3 = 10.17$

$r_4 = 3.67$

Rankings

1: T_3

2: T_1

3: T_4

4: T_2

Computer lab

2)

a) $(A^T A)\begin{bmatrix} a_1 \\ b_1 \end{bmatrix} = A^T y$

$a_1 = 2.45$

$b_1 = 1.98$

b $\frac{\partial \phi}{\partial b_2} = -2\sum_{i=1}^{5}(y_i - m_2(x_i))x_i$

$\frac{\partial \phi}{\partial b_2} = -2\sum_{i=1}^{5}(y_i - m_2(x_i))(2x_i e^{b_2 x_i})$

c $B_0 = I$

$a_2 = 1.5418$

$b_2 = 1.9978$